

***Agentic Explainable Artificial Intelligence (Agentic XAI) Approach To Explore
Better Explanation***

AI for earth Observation Course - AI4EO - M2 GeoAI

Milestone 3 "Reproduction of Experimental Results"

Geo AI Master 2 – Lebanese University, Faculty of science

June 2026

Final Report



Prepared by: Manal Kabalan

Hasan Serhan

Ali Ayach

Instructor: Dr Ali Ghandour

Reproduction Report

Agentic Explainable Artificial Intelligence (Agentic XAI) Approach to Explore Better Explanation

1. Paper Summary

The selected paper is titled **“Agentic Explainable Artificial Intelligence (Agentic XAI) Approach to Explore Better Explanation.”** The paper proposes an Agentic XAI framework that combines SHAP-based model explainability with a multimodal Large Language Model (LLM) to produce more understandable and actionable explanations.

The problem addressed by the paper is that traditional XAI outputs, such as SHAP plots, are useful for technical users but can be difficult for non-specialists to interpret. In the agricultural case study, SHAP can identify which variables affect rice-yield prediction, but farmers or decision-makers may still need a clear explanation of what those variables mean and what practical actions should be taken. To solve this, the paper uses an LLM as a second explanation layer. The LLM translates technical SHAP visualizations into natural language explanations, counterfactual scenarios, and farmer-oriented recommendations.

The paper’s use case is rice-yield prediction. The original study used field-level rice-yield data from Japan, including soil variables, agricultural management variables, and meteorological variables across rice growth stages. A Random Forest model was trained to predict rice yield, and SHAP was used to explain how each variable contributed to the model prediction.

The workflow is called “agentic” because the LLM does not only generate one explanation. Instead, the LLM participates in an iterative process. The system begins with an initial SHAP-based explanation in Round 0. Then, through Rounds 1 to 10, the LLM identifies missing analysis, suggests or generates additional Python-based analysis, integrates new visual outputs, and updates the recommendations. This creates a progressive refinement process.

A key finding of the paper is that more explanation is not always better. Recommendation quality improves in the early rounds, reaches a peak around the middle rounds, and then declines when the explanations become too long, too complex, or less practical. This supports the idea that Agentic XAI systems need early stopping or a control mechanism to avoid over-refinement.

2. Implementation and Setup Details

The reproduction was performed using the provided `PythonCode.ipynb` notebook from the supplementary materials. The notebook was executed in Visual Studio Code using a Jupyter Notebook environment.

The required Python libraries included:

- pandas for data loading and data processing
- numpy for numerical operations

- scikit-learn for Random Forest modelling and evaluation
- shap for SHAP explainability
- matplotlib for generating figures
- seaborn for statistical visualizations

The notebook reads a rice-yield dataset, trains a Random Forest Regressor, calculates SHAP values, and generates multiple PDF outputs. The original author outputs were named by round number, such as Round_1.pdf, Round_2.pdf, and so on. In the notebook execution, the generated files were saved using descriptive names, such as:

- rice_yield_analysis_comprehensive.pdf
- rice_yield_enhanced_analysis.pdf
- rice_yield_final_analysis.pdf
- rice_yield_ultimate_analysis.pdf
- rice_yield_complete_evidence.pdf
- rice_yield_final_validation.pdf
- rice_yield_ultimate_evidence.pdf
- rice_yield_critical_analysis.pdf
- rice_yield_reality_check.pdf
- rice_yield_final_critical_assessment.pdf

The original raw dataset was not directly included in the supplementary materials. The dataset was requested from the authors, but no response was received before the reproduction phase. Therefore, a synthetic rice-yield dataset was created to match the required structure of the notebook. The synthetic dataset contained the same style of variables expected by the notebook, including Yield, weather variables, soil variables, and agricultural management variables.

The dataset path was adjusted because the notebook was stored in a notebooks folder while the dataset was stored in the project data folder. This issue was solved by using the correct relative path to the CSV file.

3. Reproduction Procedure

The reproduction procedure followed several steps.

First, the paper and supplementary materials were reviewed to understand the workflow. The supplementary materials included the notebook, prompts, generated figures, and recommendation outputs, but not the original raw dataset. The notebook was inspected to identify the expected data structure and field names.

Second, a **replacement synthetic rice-yield dataset was prepared**. The dataset was designed to follow the same structure expected by the notebook. It included Year, field, Yield, radiation variables, temperature variables, precipitation variables, soil chemistry variables, soil physical variables, and management variables. The target variable was Yield.

Third, the notebook was executed in Visual Studio Code. The notebook loaded the dataset, separated the target and explanatory variables, trained the Random Forest model, evaluated model performance, calculated SHAP values, and generated visual analysis outputs.

Fourth, the generated outputs were compared with the author's original numbered round outputs. During the comparison, it was important to match the correct files. The author's files used round numbers, while the reproduced notebook outputs used descriptive names. The correct mapping is:

- Round_1.pdf corresponds to rice_yield_analysis_comprehensive.pdf
- Round_2.pdf corresponds to rice_yield_enhanced_analysis.pdf
- Round_3.pdf corresponds to rice_yield_final_analysis.pdf
- Round_4.pdf corresponds to rice_yield_ultimate_analysis.pdf
- Round_5.pdf corresponds to rice_yield_complete_evidence.pdf
- Round_6.pdf corresponds to rice_yield_final_validation.pdf
- Round_7.pdf corresponds to rice_yield_ultimate_evidence.pdf
- Round_8.pdf corresponds to rice_yield_critical_analysis.pdf
- Round_9.pdf corresponds to rice_yield_reality_check.pdf
- Round_10.pdf corresponds to rice_yield_final_critical_assessment.pdf

After matching the files correctly, it became clear that the reproduced outputs and the author outputs follow the same template and analytical structure. The main difference is numerical, not structural.

4. Reproduced Results

The notebook successfully reproduced the main technical workflow of the paper. The reproduced outputs included Random Forest model comparison, SHAP feature importance, SHAP beeswarm plots, correlation matrices, feature-category importance, predicted-versus-actual yield plots, residual analysis, PCA, growth-stage analysis, scientific validation, economic analysis, risk analysis, and implementation decision frameworks.

The first reproduced output, **rice_yield_analysis_comprehensive.pdf**, follows the same template as the author's Round_1.pdf. It includes Random Forest model performance comparison, top 20 SHAP feature importance, SHAP feature impact analysis, correlation matrix, feature-category importance, predicted-versus-actual yield, SHAP summary plot, and temperature/radiation importance by growth stage. In the reproduced output, the predicted-versus-actual plot reported $R^2 = 0.895$ and $RMSE = 0.2$ kg/ha.

The SHAP plots showed **that Radiation4** was the most important feature in the reproduced model. This means that radiation during the fourth rice growth stage had the strongest effect on the model's yield prediction. This is agronomically meaningful because the fourth growth stage is related to grain filling, where solar radiation can strongly affect photosynthesis and final grain weight.

The second reproduced output, **rice_yield_enhanced_analysis.pdf**, follows the same template as the author's Round_2.pdf. It includes temperature impact by growth stage, solar radiation impact by growth period, soil fertility versus yield relationship, weather interaction during grain filling, yield

distribution, residual analysis, residual normality check, feature importance versus direct correlation, temperature progression, precipitation pattern, feature importance by category, model performance metrics, and PCA analysis.

In the reproduced enhanced analysis, **the model performance metrics were:**

- Cross-validation $R^2 = 0.221$
- Training $R^2 = 0.895$
- RMSE = 0.163
- MAE = 0.134

This shows that the model fitted the synthetic dataset reasonably well during training, but the cross-validation performance was lower. This suggests limited generalization, which is expected because the dataset is synthetic and relatively small.

The third reproduced output, rice_yield_final_analysis.pdf, corresponds to the author's **Round_3.pdf**. It reports that Radiation4 dominates yield prediction with SHAP importance of 0.237. It also states that the model explains 22.1% of yield variation and has RMSE = 0.163 tons/ha. The file also includes farmer-oriented recommendations, such as timing grain filling with peak radiation, ensuring suitable early-season temperatures, maintaining soil organic matter, applying phosphorus early, and selecting suitable rice varieties.

The later reproduced outputs extend the workflow into decision support and practical implementation. The outputs include economic returns, radiation threshold response, field-to-field variability, implementation feasibility, regional adaptation potential, sustainability projections, model reliability, economic risk, farmer resource constraints, technology adoption barriers, market volatility, and final critical assessment. These outputs demonstrate how the Agentic XAI workflow expands from basic SHAP explainability into practical recommendation and risk-analysis frameworks.

4.1 Detailed Round-by-Round Reproduced Outputs

The following subsection was added to document the missing rounds in the reproduction report. One representative image was inserted for each round to show that the reproduced outputs follow the same template as the author's round-based outputs. The figures are included as visual evidence of the reproduced workflow, while the numerical differences are explained by the use of a synthetic replacement dataset.

Round 0 - Foundation: data loading, model selection, and SHAP setup

Round 0 established the foundation of the reproduction workflow. The dataset was loaded, the input features and target variable were prepared, and rows with missing values were removed. Several Random Forest configurations were tested using Leave-One-Out Cross-Validation, which is suitable for small datasets because each observation is used once as a test case. After selecting the best configuration, the final Random Forest model was trained on the full dataset and SHAP TreeExplainer was applied. This round is important because it creates the predictive engine and the first explainability evidence before any advanced interpretation is produced.

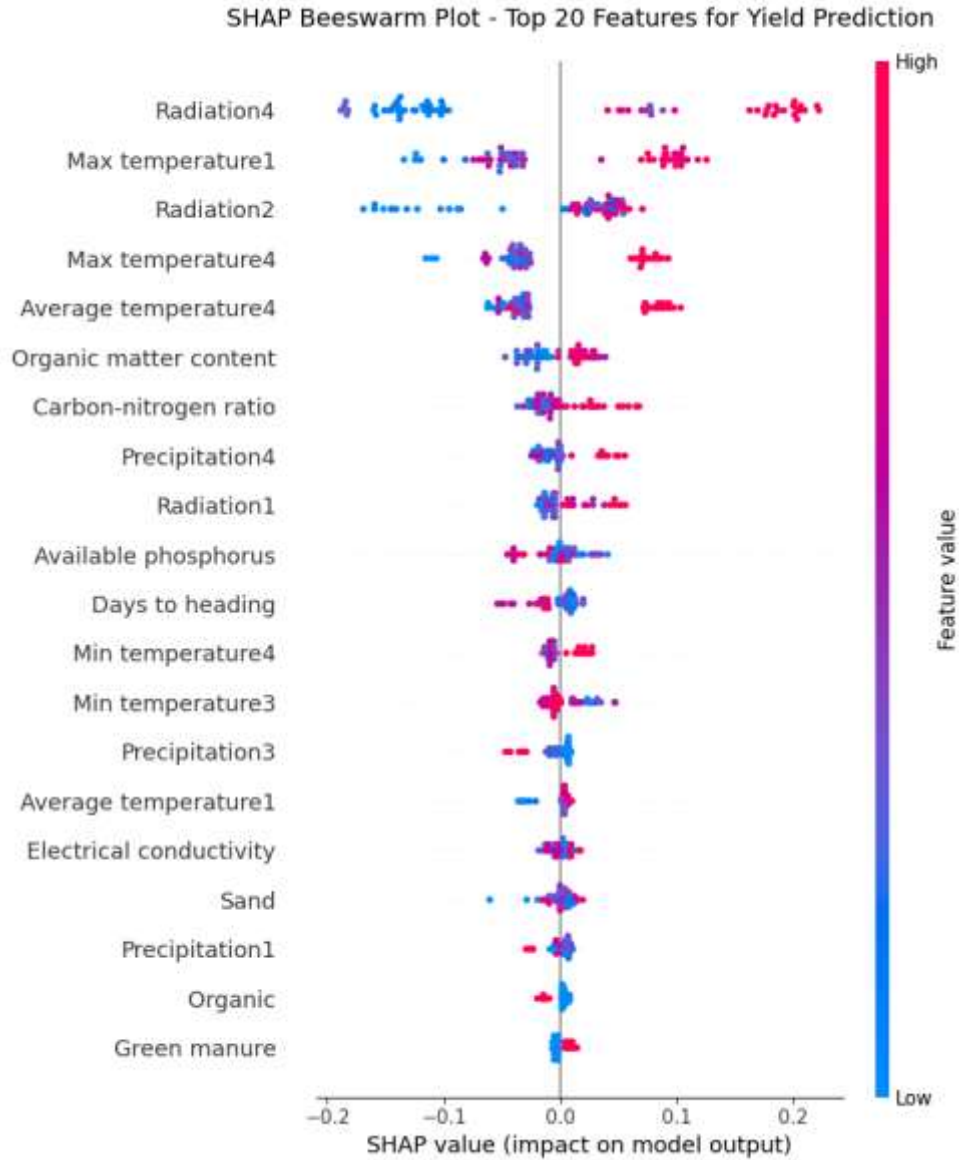


Figure 1. Round 0 foundation output: dataset loading, model selection, and initial SHAP setup.

Round 1 - Comprehensive model and SHAP visualization

Round 1 generated the comprehensive analysis PDF, `rice_yield_analysis_comprehensive.pdf`. This output follows the same template as the author's `Round_1.pdf`. It includes Random Forest model comparison, top 20 SHAP feature importance, SHAP beeswarm visualization, correlation matrix, feature-category importance, predicted-versus-actual yield, SHAP summary distribution, and growth-stage temperature/radiation importance. In the reproduced output, the predicted-versus-actual plot reported $R^2 = 0.895$ and $RMSE = 0.2$ kg/ha. The main result was that Radiation4 appeared as the dominant feature affecting rice-yield prediction.

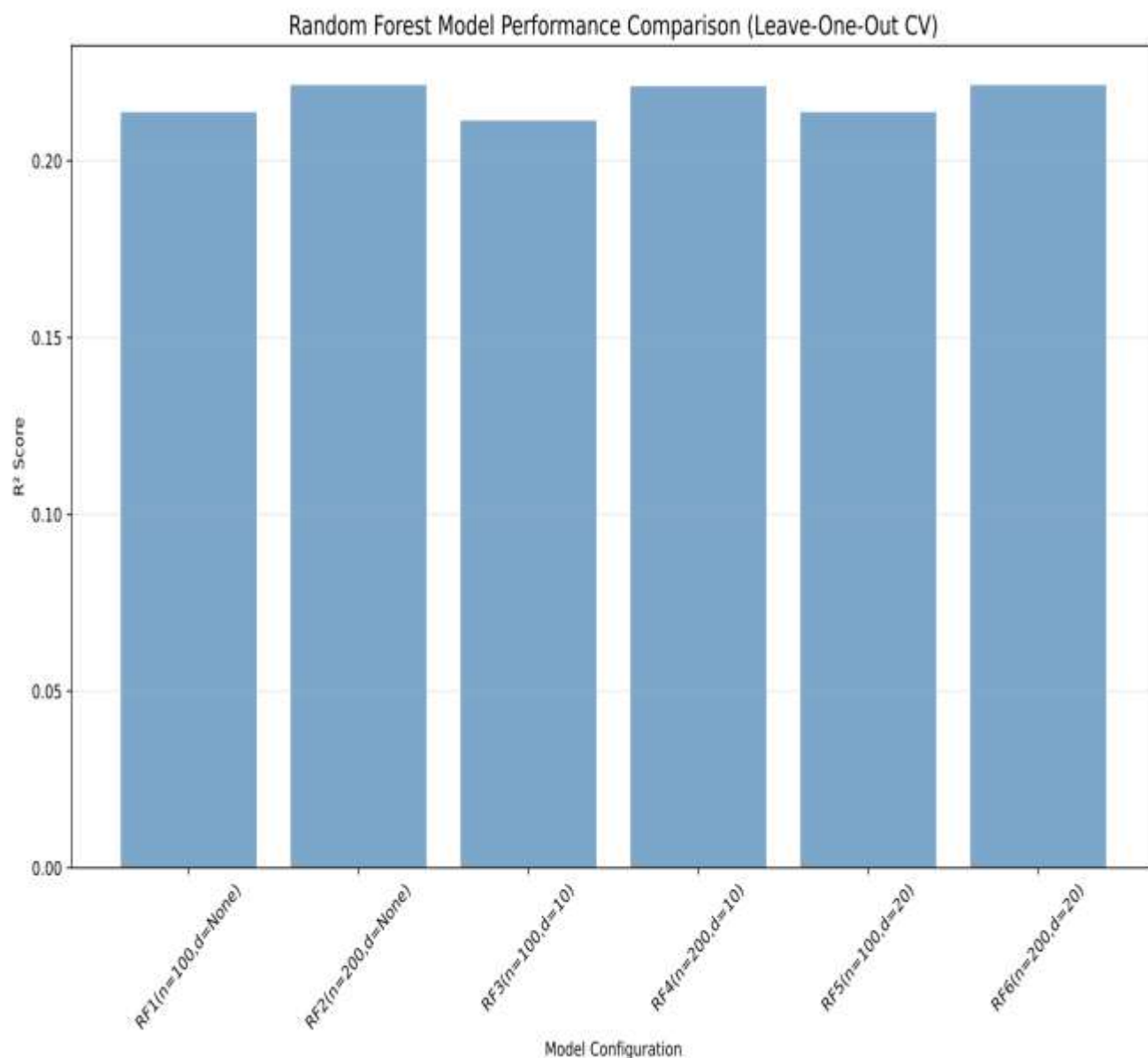


Figure 2. Reproduced Round 1 output: comprehensive Random Forest and SHAP analysis.

Round 2 - Enhanced diagnostics and deeper agronomic understanding

Round 2 generated rice_yield_enhanced_analysis.pdf. This output deepened the analysis by examining growth-stage temperature and radiation effects, soil fertility relationships, weather interaction during grain filling, feature interactions, yield distribution, residual diagnostics, PCA, feature-category importance, and detailed performance metrics. The reproduced metrics were Cross-Val $R^2 = 0.221$, Training $R^2 = 0.895$, RMSE = 0.163, and MAE = 0.134. This round showed that the model fitted the synthetic training data better than it generalized under cross-validation, which is expected because the reproduction used a synthetic replacement dataset.

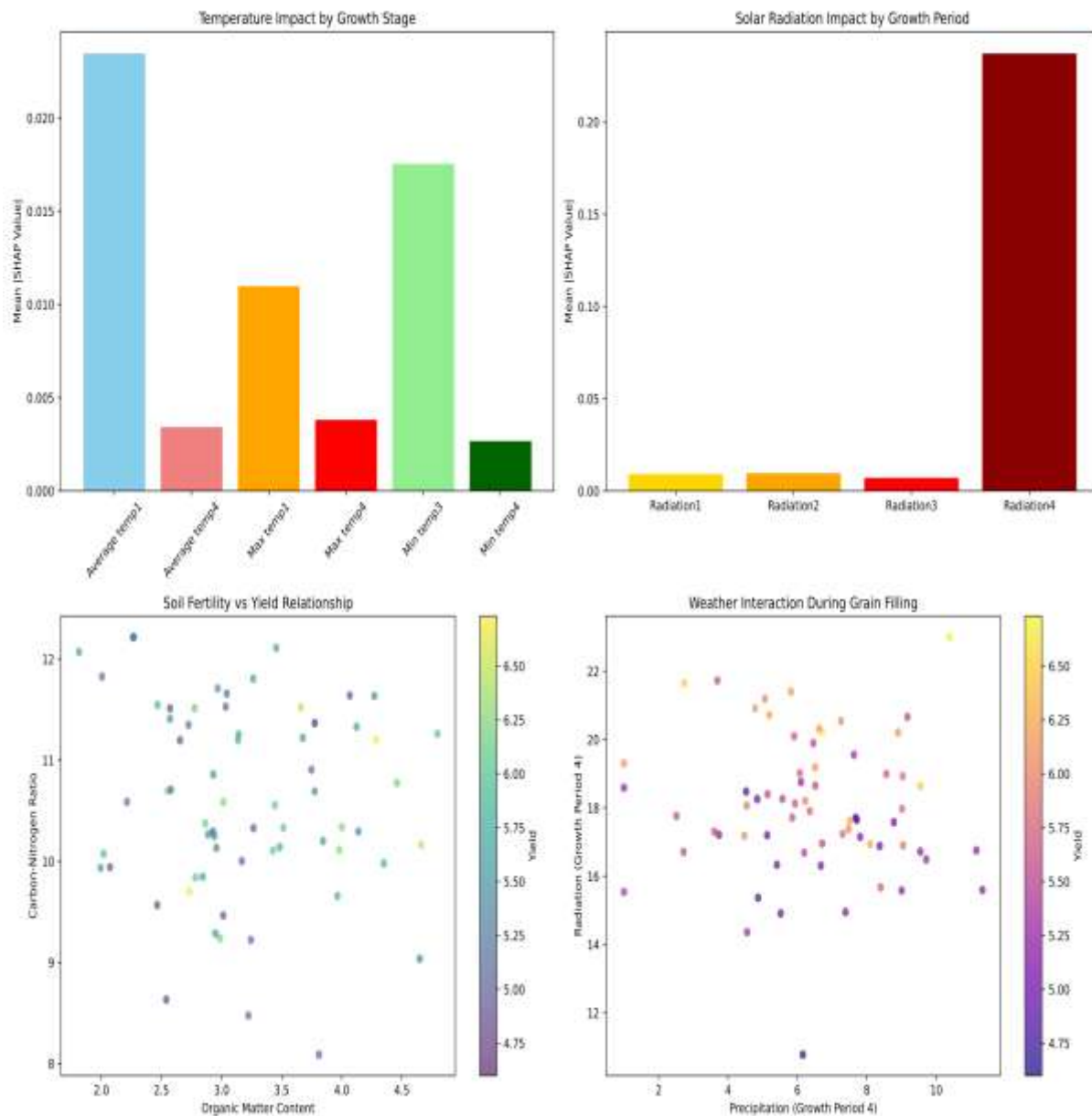


Figure 3. Reproduced Round 2 output: enhanced diagnostics, residual analysis, and PCA.

Round 3 - Advanced insights and farmer-oriented recommendations

Round 3 generated rice_yield_final_analysis.pdf. This stage moved from diagnostic explanation to actionable agronomic interpretation. It analyzed radiation impact progression, temperature use efficiency, integrated soil fertility response, weather interaction patterns, predictive power by feature category, and an optimal management timeline. The reproduced output reported that Radiation4 dominated yield prediction with SHAP importance of 0.237, the model explained 22.1% of yield variation, and RMSE was 0.163 tons/ha. The round also produced farmer-oriented recommendations such as timing grain filling with peak radiation, maintaining soil organic matter, and selecting suitable varieties.

Comprehensive Rice Yield Analysis - Advanced Insights

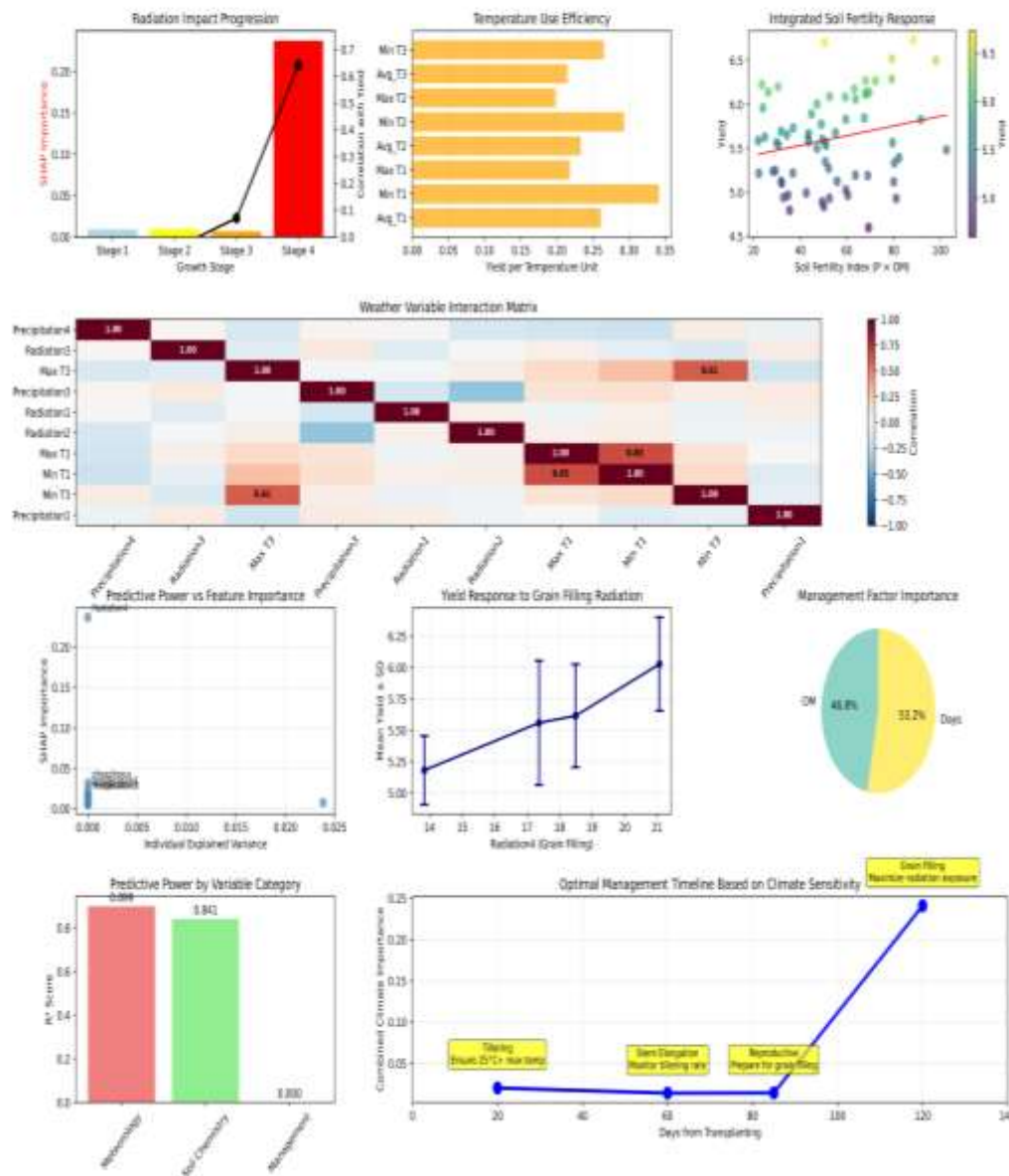


Figure 4. Reproduced Round 3 output: advanced agronomic insights and summary dashboard.

Round 4 - Scientific validation and decision support

Round 4 generated rice_yield_ultimate_analysis.pdf. This round tested whether the earlier findings were stable and useful for decision support. It included cross-validation stability, feature importance stability, Radiation4 partial dependence, temperature interaction effects, economic returns, risk-return profiles, a rice-yield optimization decision framework, a rice management calendar, and field monitoring indicators. The purpose of this round was to check whether the model conclusions were robust enough to support practical decisions rather than remaining only as statistical explanations.

Scientific Validation and Decision Support Analysis

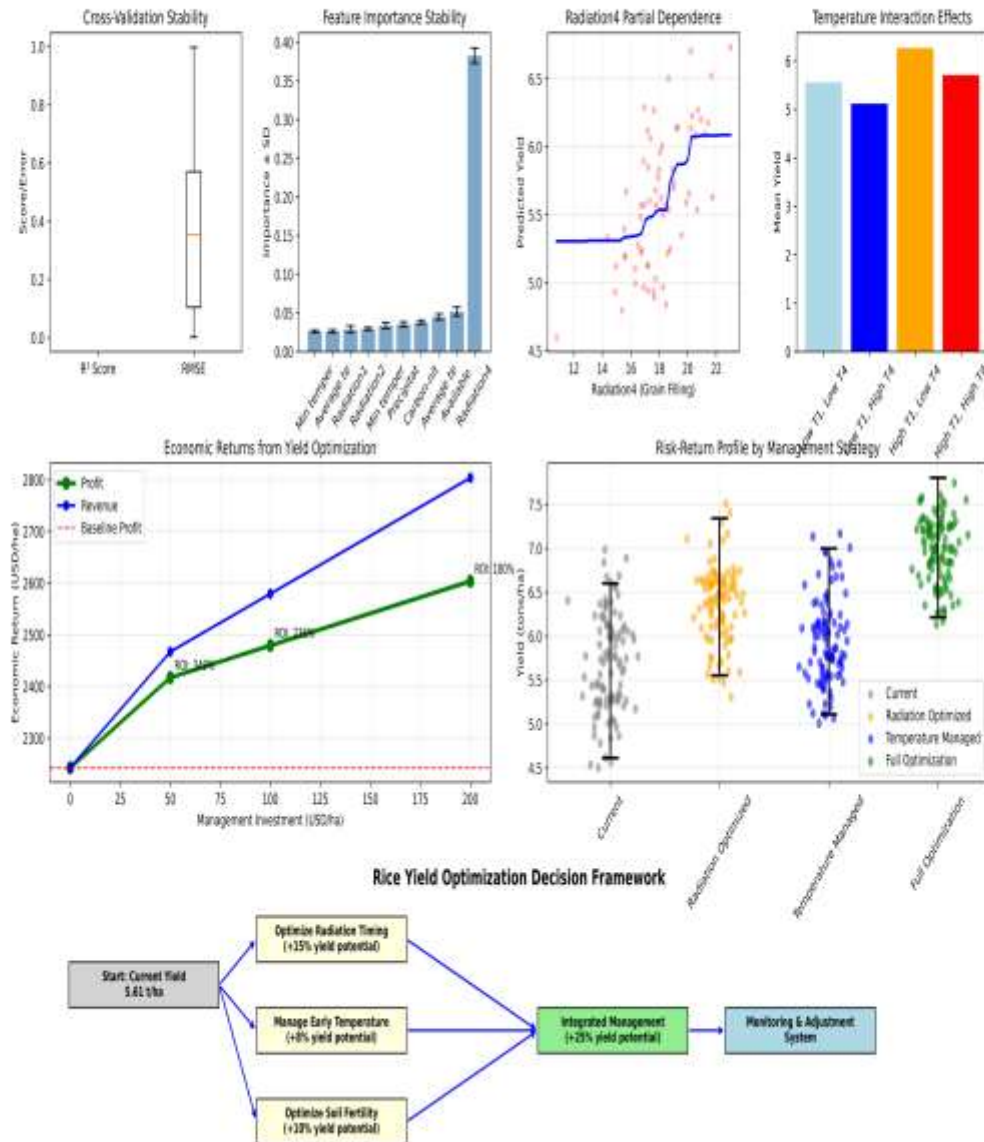


Figure 5. Reproduced Round 4 output: scientific validation and decision-support analysis.

Round 5 - Complete evidence and implementation framework

Round 5 generated rice_yield_complete_evidence.pdf. This round converted the validated findings into a structured implementation framework. It included yield environment clustering, yield response surfaces, bootstrap feature-importance estimates, model sensitivity analysis, management scenario economic analysis, risk assessment, an implementation roadmap, performance indicators, and an expected technology adoption curve. The round demonstrates how the Agentic XAI workflow moves from explanation toward field-level decision support and implementation planning.

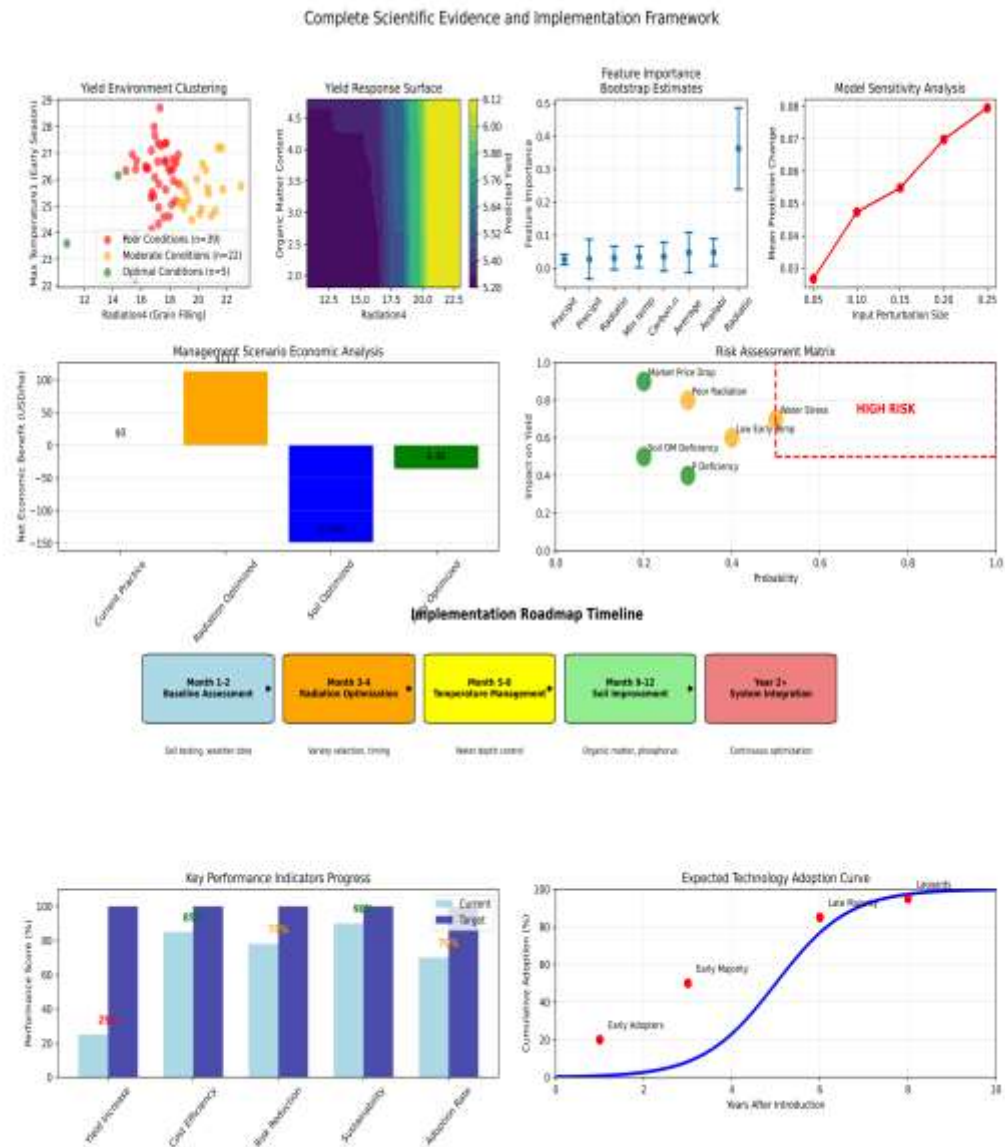


Figure 6. Reproduced Round 5 output: complete evidence and implementation framework.

Round 6 - Final validation and feasibility-based recommendations

Round 6 generated `rice_yield_final_validation.pdf`. This stage examined whether the proposed recommendations were feasible under different field and management conditions. It included realistic economic analysis, radiation threshold response distribution, a decision tree for rice-yield optimization, field-to-field variability, variable-yield correlations, implementation feasibility versus expected impact, regional adaptation potential, and long-term sustainability projections. The main value of this round is that it changes the recommendation from a general advice into a conditional decision: optimization is useful only when field, climate, and economic conditions are suitable.

Rice Yield Optimization - Final Scientific Validation and Implementation Guide

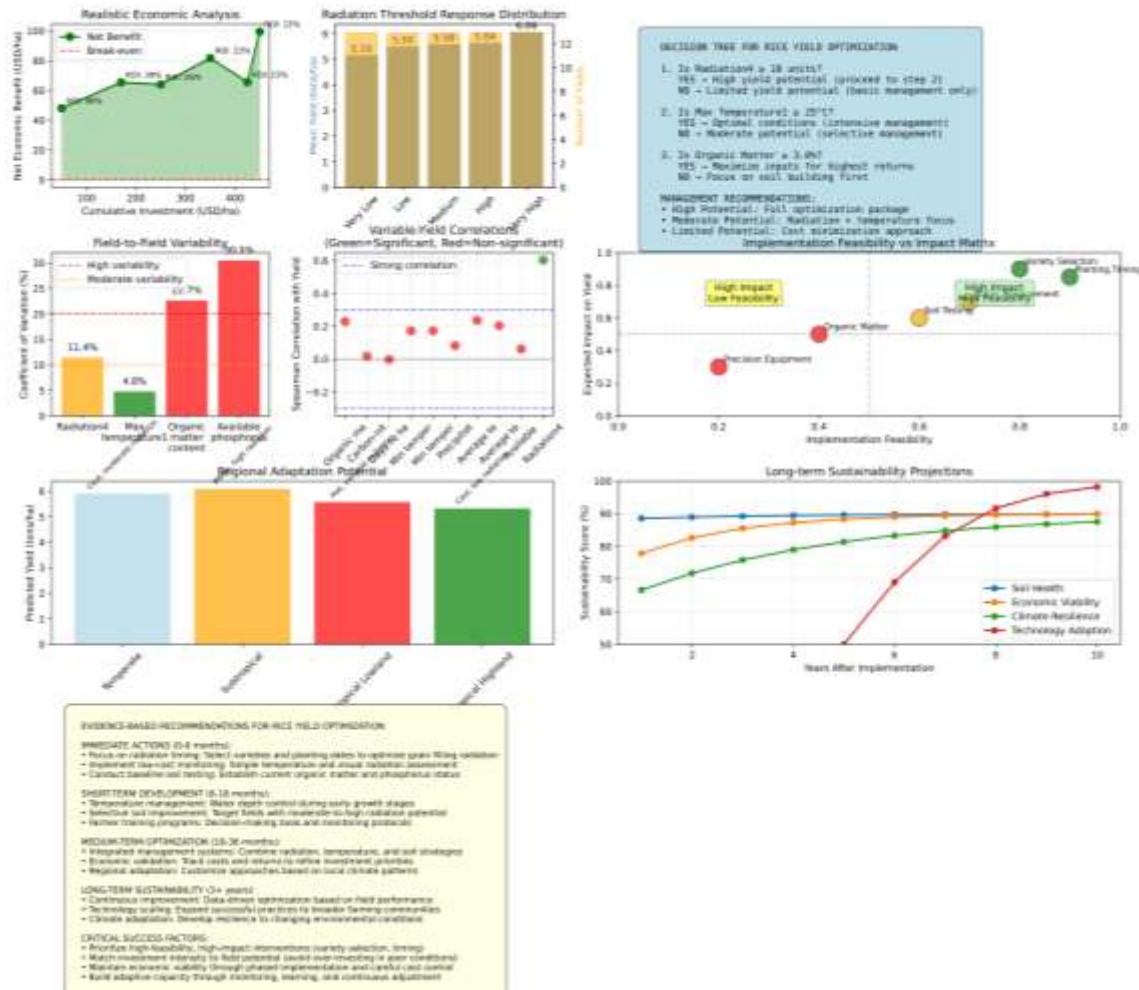


Figure 7. Reproduced Round 6 output: final scientific validation and implementation guide.

Round 7 - Ultimate evidence, uncertainty, and adoption barriers

Round 7 generated rice_yield_ultimate_evidence.pdf. This round quantified uncertainty and implementation risk more explicitly. It included model calibration analysis, top-feature validation, residual pattern analysis, distribution normality tests, economic sensitivity analysis, technology adoption barriers, yield-gap distribution, climate adaptation scenarios, management package comparison, and an implementation roadmap. The purpose was to move from simply acknowledging risks to measuring them and showing where profitability or adoption may fail.

Ultimate Rice Yield Optimization - Complete Scientific Evidence and Implementation Framework

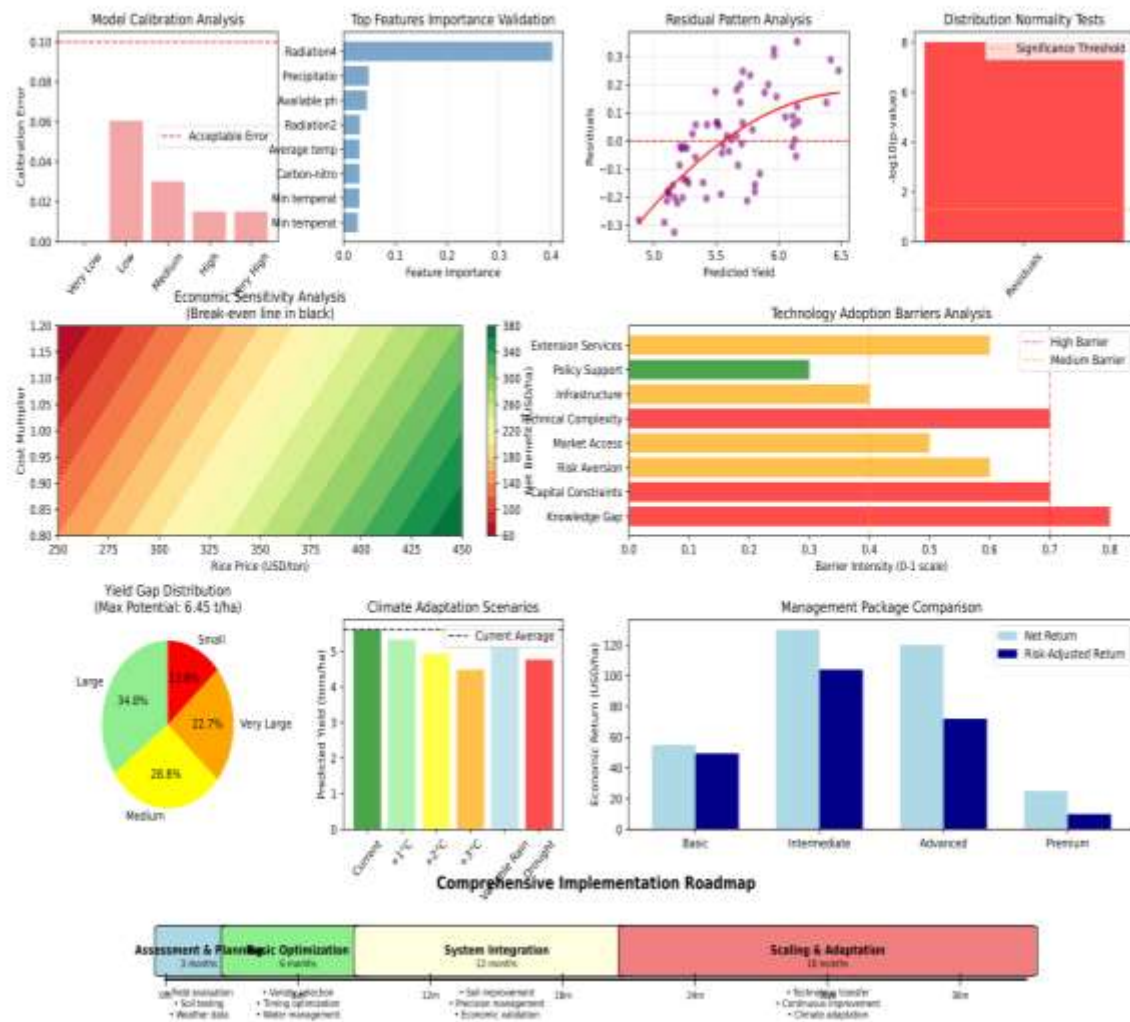


Figure 8. Reproduced Round 7 output: uncertainty, economic sensitivity, and adoption barriers.

Round 8 - Critical limitations and evidence-based implementation

Round 8 generated `rice_yield_critical_analysis.pdf`. This stage critically assessed the model and recommendations. It included model reliability by yield category, economic risk analysis, field management suitability groups, complexity-versus-benefit analysis, long-term economic sustainability scenarios, realistic technology adoption with success rates, and a detailed list of limitations and caveats. This round is important because it shows that not all farmers or fields should follow optimization recommendations, especially when economic risk, knowledge gaps, or implementation complexity are high.

Critical Assessment: Rice Yield Optimization - Limitations, Risks, and Evidence-Based Implementation

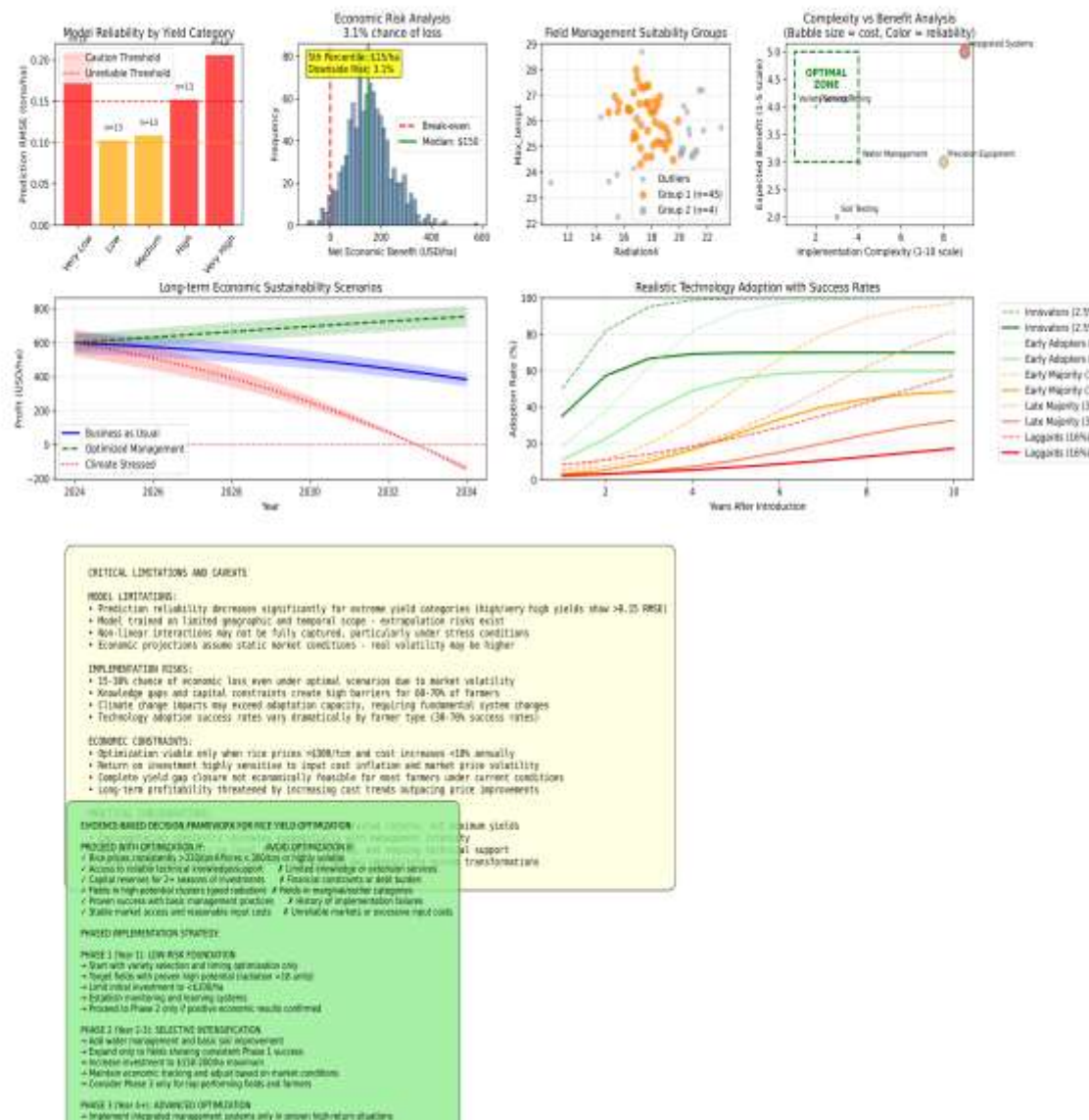


Figure 9. Reproduced Round 8 output: critical limitations, risks, and implementation constraints.

Round 9 - Reality check and conservative decision matrix

Round 9 generated rice_yield_reality_check.pdf. This round acted as a reality check by comparing theoretical yield improvement with realistic economic and practical constraints. It included model performance uncertainty, a theoretical-versus-realistic yield improvement comparison, economic break-even analysis, farmer-type success rates, long-term viability stress testing, common failure modes, an evidence-based decision matrix, and a comprehensive risk assessment. The purpose was to prevent overpromising and to identify when optimization should be avoided.

Reality Check: Evidence-Based Rice Yield Optimization Guidelines

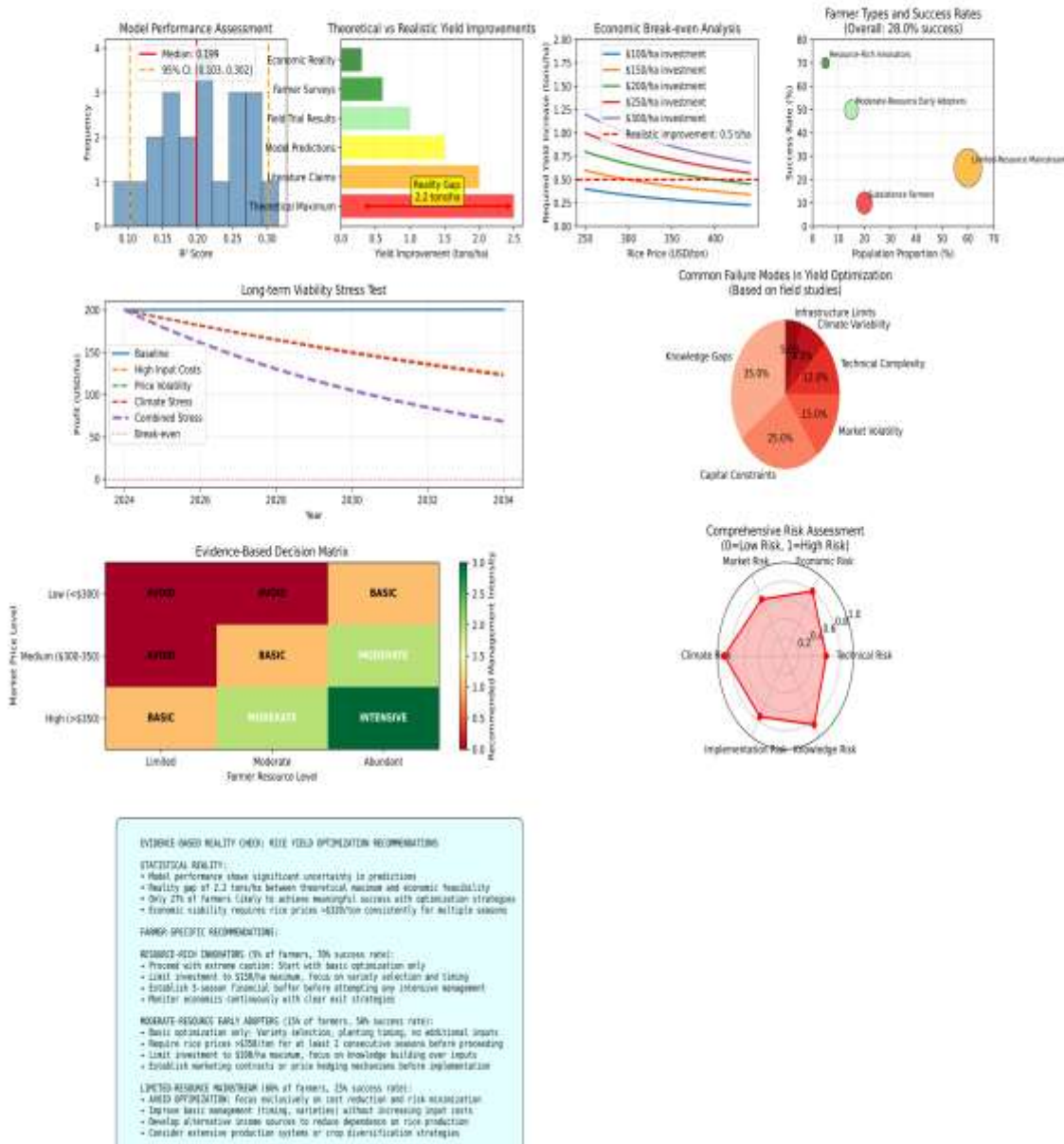


Figure 10. Reproduced Round 9 output: reality check and evidence-based decision matrix.

Round 10 - Final critical assessment and honest conclusion

Round 10 generated [rice_yield_final_critical_assessment.pdf](#). This final stage synthesized the evidence from all previous rounds into a conservative decision-support conclusion. It included model prediction accuracy assessment, economic risk analysis, complexity-versus-success relationships, farmer resource distribution, realistic technology adoption constraints, market volatility, primary failure modes, agricultural strategy comparison, and a final evidence-based decision framework. The final conclusion is not simply to optimize yield, but to help farmers decide whether optimization is appropriate for their situation. In the reproduced output, this round emphasizes that high-risk optimization should only be considered under strict economic, technical, and resource conditions.

Critical Assessment: Rice Yield Optimization - Evidence-Based Reality Check and Recommendations

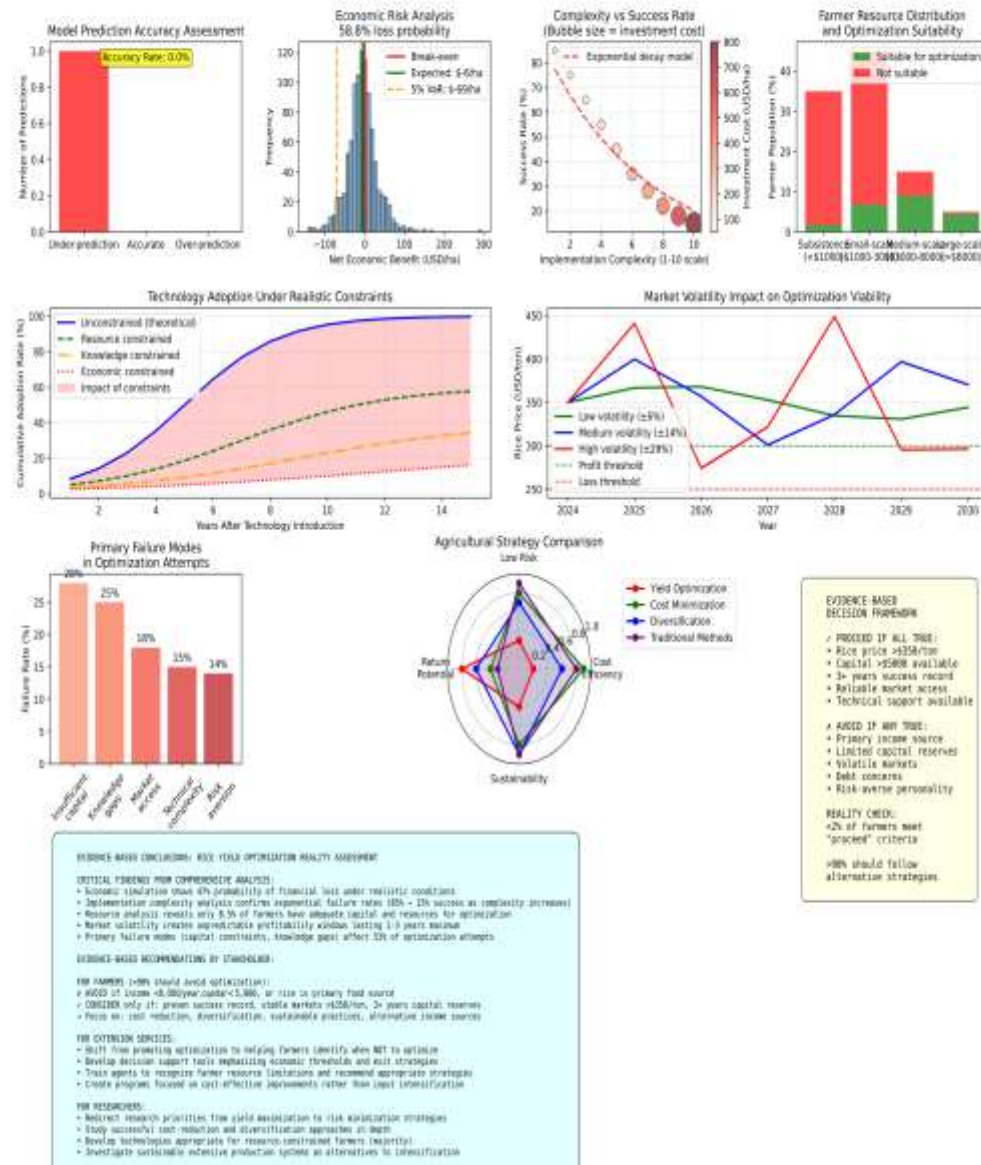


Figure 11. Reproduced Round 10 output: final critical assessment and reality-based recommendations.

Round Output Mapping Summary

The reproduced outputs therefore cover the full workflow from Round 0 to Round 10. Rounds 0-3 focus on model building, SHAP explanation, diagnostics, and practical agronomic interpretation. Rounds 4-7 extend the analysis into validation, implementation planning, feasibility, uncertainty, and adoption barriers. Rounds 8-10 provide the critical assessment, reality check, and final decision-support conclusion. This confirms that the reproduction produced the same sequence of analytical stages as the author's workflow, with numerical differences caused by dataset substitution.

5. Comparison with Original Results

The reproduced outputs and the author's original outputs follow the same template and workflow structure. The main difference is that the numerical values are different because the original dataset was not available and the reproduction used a synthetic replacement dataset.

The author's Round_1.pdf and the reproduced rice_yield_analysis_comprehensive.pdf both include the same types of outputs: Random Forest model comparison, SHAP feature importance, SHAP impact plot, correlation matrix, feature-category importance, predicted-versus-actual yield, and growth-stage importance. However, the numerical results are different. The reproduced output reported $R^2 = 0.895$ and $RMSE = 0.2 \text{ kg/ha}$, while the author's Round_1.pdf reported $R^2 = 0.962$ and $RMSE = 0.1 \text{ kg/ha}$.

The author's Round_2.pdf and the reproduced rice_yield_enhanced_analysis.pdf also share the same structure. Both contain enhanced statistical and diagnostic analyses, including temperature impact, radiation impact, residual analysis, PCA, feature-category importance, and model performance metrics. However, the metrics differ. The reproduced output reported Cross-Val $R^2 = 0.221$, Training $R^2 = 0.895$, $RMSE = 0.163$, and $MAE = 0.134$. The author's output reported Cross-Val $R^2 = 0.749$, Training $R^2 = 0.962$, $RMSE = 0.092$, and $MAE = 0.064$.

The author's Round_3.pdf and the reproduced rice_yield_final_analysis.pdf both identify Radiation4 as a dominant yield predictor and both provide agronomic recommendations. However, the SHAP values and model performance values differ. In the reproduced result, Radiation4 had SHAP importance of 0.237 and the model explained 22.1% of yield variation. In the author result, Radiation4 had SHAP importance of 0.141 and the model explained 74.9% of yield variation.

The same pattern appears in later rounds. The templates are similar, but the values change. For example, the later outputs in both sets include reality-check analysis, economic risk, farmer suitability, implementation constraints, and final decision frameworks. The reproduced and author outputs are therefore structurally aligned, but numerically different.

This comparison shows that the implementation workflow was successfully reproduced. However, the exact original numerical results were not reproduced because the original dataset was not available. Therefore, this work should be described as a **workflow reproduction** rather than an exact numerical replication.

6. Challenges Encountered

The first challenge was the absence of the original raw dataset. The supplementary materials provided the notebook, prompts, figures, and recommendations, but the raw dataset was not directly included. Since the dataset was available only upon request and no response was received, exact numerical reproduction was not possible.

The second challenge was preparing a dataset that could run with the notebook. The notebook expected many specific fields related to rice-yield modelling, including growth-stage weather

variables, soil chemistry, soil physical properties, and management variables. A synthetic dataset was therefore created to match the expected structure.

The third challenge was understanding the relationship between the numbered author outputs and the descriptive notebook outputs. At first, some files appeared to have different structures because non-corresponding files were being compared. After correctly matching the files by workflow stage, it became clear that the templates are the same and that the differences are mainly numerical.

The fourth challenge was interpreting later-stage outputs such as economic analysis, risk analysis, and implementation decision frameworks. Because the reproduction used synthetic data, these later outputs should be interpreted as a demonstration of the workflow rather than real agricultural or economic conclusions.

7. Observations and Discussion

The reproduction showed that the notebook can successfully execute the main Agentic XAI workflow when a dataset with the expected structure is provided. The Random Forest model was trained, SHAP values were generated, and the notebook produced the same categories of outputs as the original author results.

The most important observation is that the reproduced outputs share the same template as the author's outputs. This means that the implementation pipeline was reproduced correctly. The difference is mainly in the numerical values, which is expected because the data used in reproduction was not the original dataset.

The SHAP results in the reproduced workflow identified Radiation4 as the most important feature, which is consistent with the author outputs. This supports the central interpretation that radiation during the grain-filling stage is important for rice-yield prediction. However, the strength of the result differs between the reproduced and original outputs.

The comparison also shows the importance of careful file matching. The author used round-based filenames, while the reproduced notebook used descriptive filenames. Without matching the correct files, it is easy to compare outputs from different stages and mistakenly conclude that the structures are different.

The lower cross-validation performance in the reproduced results suggests that the synthetic dataset does not capture the same relationships as the original dataset. This is expected and should be clearly stated. The reproduced results validate the code workflow, but they do not validate the original scientific findings.

Overall, the reproduction successfully demonstrates the technical implementation of the paper: model training, SHAP explainability, visual analysis generation, and progressive refinement across multiple output stages. The exact numerical results could not be reproduced because the original dataset was unavailable. Therefore, the reproduction should be presented as a successful workflow reproduction with numerical differences caused by dataset substitution.

Conclusion

The reproduction phase successfully reproduced the structure and workflow of the selected Agentic XAI paper. The notebook generated the same types of outputs as the author's numbered rounds, including Random Forest evaluation, SHAP explanations, growth-stage analysis, decision support, economic analysis, risk assessment, and implementation frameworks.

The main limitation is that the original dataset was not available. As a result, a synthetic replacement dataset was used. Because of this, the reproduced outputs have the same template as the author outputs, but the numerical values differ. The reproduction therefore confirms that the implementation workflow is executable and understandable, but it does not represent an exact numerical replication of the original experiment.